ENGINEERING MATHEMATICS – I

Course Syllabus & Structure (M1)

Course Code:	BAS-103 / M1	**Total Contact Hours:**	40 Hours
Category:	Basic Science Course	**Evaluation Scheme:**	Theory (L-T-P: 3-1-0)

Module / Unit Content	Contact Hours
Unit 1: Matrices	**8 Hours**
Elementary transformations, Inverse of a matrix, Rank of a matrix, Solution of system of linear equations, Characteristic equation, Cayley-Hamilton Theorem and its applications, Linear Dependence and Independence of vectors, Eigenvalues and Eigenvectors, Complex Matrices, Hermitian, Skew-Hermitian and Unitary Matrices, Applications to Engineering problems.	
Unit 2: Differential Calculus – I	**8 Hours**
Successive Differentiation (n^{th} order derivatives), Leibniz's theorem, Curve tracing, Partial derivatives, Euler's Theorem for homogeneous functions, Total derivative, Change of variables.	
Unit 3: Differential Calculus – II	**8 Hours**
Expansion of functions by Taylor's and Maclaurin's theorems for functions of one and two variables, Maxima and Minima of functions of several variables, Lagrange's method of multipliers, Jacobians, Approximation of errors.	
Unit 4: Multiple Integration	**8 Hours**
Double integrals, Triple integrals, Change of order of integration, Change of variables, Beta and Gamma functions and their properties, Dirichlet's integral and its applications to area and volume, Liouville's extensions of Dirichlet's integral.	
Unit 5: Vector Calculus	**8 Hours**
Vector Differentiation: Gradient, Curl and Divergence and their physical interpretations, Directional derivatives. **Vector Integration:** Line integrals, Surface integrals, Volume integrals, Gauss's Divergence theorem, Green's theorem, and Stokes' theorem (without proof) and their engineering applications.	